

INT335: DESIGN THINKING

High-Yield Exam Revision Handbook

Target Curriculum:

B.Tech Computer Science & Engineering | Semester 3
Lovely Professional University

Revision Handbook Highlights:

- **High-Yield Concept Cards:** Core theoretical models, psychological axioms, and cognitive biases.
- **Comparison Matrices:** Qualitative vs. Quantitative, Lo-Fi vs. Hi-Fi, Usability vs. CX, Refactoring vs. Re-design.
- **Formulas & Metrics:** Screening matrix scoring, priority formula $P = F \times S \times C$, WCAG contrast ratios.
- **Landmark Case Studies:** OXO Good Grips, M-Pesa, Jaipur Foot, Juicero, Google Glass, and Ford Edsel.

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CHAPTER 1

Unit 1: Foundations of Learning, Creativity and Design Thinking

Unit 1 Essential Summary

- **Design Thinking:** A non-linear, human-centred, iterative problem-solving methodology that starts with human empathy rather than technology.
- **Wicked Problems (Rittel & Webber, 1973):** Ill-defined social/systemic problems with no definitive formulation, no immediate test, and no single correct answer.
- **Three Lenses of Innovation (IDEO):** Desirability (Human), Feasibility (Technical), and Viability (Business).

1.1 Cognitive Dynamics & Learning Models

Kolb's Experiential Learning Cycle

Concrete Experience → Reflective Observation → Abstract Conceptualization → Active Experimentation

- **Focused vs. Diffuse Thinking (Dr. Barbara Oakley):**
 - *Focused Mode:* Concentrated attention, analytical step-by-step logic, error detection. (Trap: tunnel vision).
 - *Diffuse Mode:* Relaxed resting-state default mode network; unconscious incubation, unexpected creative synthesis.
- **Bloom's Revised Taxonomy Hierarchy:** Remember → Understand → Apply → Analyze → Evaluate → **Create** (Design Thinking operates at the peak: Create).
- **VARK Modalities:** Visual (V), Aural (A), Read/Write (R), Kinesthetic (K).
- **Functional Fixedness (Karl Duncker, 1945):** Cognitive bias where an object is perceived only in terms of its traditional use (e.g., Duncker's Candle Problem where the matchbox must be seen as a candle shelf).

1.2 Creativity vs. Invention vs. Innovation

1.3 Milestones & Landmark Case Studies

- **1919 (Bauhaus):** Walter Gropius unites form and functional manufacturing.
- **1969 (Herbert Simon):** *The Sciences of the Artificial* — Design transforms existing situations into preferred ones.
- **1987 (Peter Rowe):** First book explicitly titled *Design Thinking*.

Term	Core Definition	Benchmark Example
Creativity	Psychological capacity to generate novel and contextually appropriate ideas.	Conceptualizing low-water sanitation methods.
Invention	Engineering a technically new device, process, or material (patentable).	Developing a micro-filtration membrane.
Innovation	Implementing, scaling, and driving commercial/social adoption of a solution.	Scaling low-cost drinking water kiosks in rural communities.

Table 1.1: Distinction Between Creativity, Invention, and Innovation.

- **1991 (IDEO):** Commercialization of Human-Centred Design (Inspiration → Ideation → Implementation).
- **2005 (Stanford d.school):** 5-Stage Model (Empathize → Define → Ideate → Prototype → Test).

Global Case Study Takeaways

- **OXO Good Grips (1990):** Universal design (thick ribbed rubber handle) created for arthritis patients became a dominant consumer hit.
- **M-Pesa (Kenya, 2007):** SMS-based micro-banking using local corner agents democratized banking for unbanked populations.
- **Jaipur Foot (India):** Low-cost waterproof prosthetic designed for local cultural habits (squatting, barefoot farming).
- **Juicero Failure (2017):** \$700 Wi-Fi juice press collapsed because packets were easily hand-squeezed; over-engineered a non-existent need.

CHAPTER 2

Unit 2: Empathy, Observation and Problem Identification

Unit 2 Essential Summary

- **User Empathy:** Stepping into the user’s lived experience to discover motivations, emotions, and hidden needs.
- **Computational Empathy:** Designing software that adapts to human context, cognitive limits, emotional distress, and accessibility.
- **Need vs. Solution:** Needs are functional/emotional outcomes (e.g., “timely arrival updates”); solutions are specific technical implementations (e.g., “mobile GPS tracker”).

2.1 Research Methods & The AEIOU Framework

Dimension	Qualitative Research	Quantitative Research
Focus	Deep “Why” and “How” behind behaviors.	Measurable “How many” and “How much”.
Methods	Semi-structured interviews, diary logs, observation.	Surveys ($N \geq 100$), clickstream telemetry, A/B tests.
Sample	Small, purposive sample ($N = 5$ to 15).	Large, statistically representative sample.

Table 2.1: Qualitative vs. Quantitative Research Comparison.

The 5 AEIOU Field Observation Lenses

- **A — Activities:** Purposeful actions and workflows people perform.
- **E — Environments:** Physical/digital context (lighting, noise, clutter, connectivity).
- **I — Interactions:** Exchanges between person-person, person-device, or person-system.
- **O — Objects:** Physical/digital tools, improvised cheat-sheets, sticky notes, keycards.
- **U — Users:** The individuals observed, their roles, behaviors, and relationships.

2.2 Inquiry Tactics, Needs, and Personas

- **Interviewing Rules:** Use open-ended narrative questions (*“Tell me about the last time you...”*); avoid leading bias (*avoid “Don’t you love our fast app?”*); use Critical Incidents (extreme stories of success/failure); apply Laddering (5 Whys) to uncover root emotional drivers.
- **Explicit vs. Implicit Needs:**
 - *Explicit Needs:* Directly stated requirements (e.g., “I need an export button”).
 - *Implicit (Latent) Needs:* Unspoken desires inferred through workarounds and hesitation (e.g., need for audit reassurance and data security).

- **Pain Point Classes:** Time Pain (delays), Process Pain (fragmented apps), Financial Pain (hidden fees), Emotional Pain (anxiety/confusion).
- **User Personas:** Research-grounded archetypes representing goals, behaviors, frustrations, and context.

2.3 Empathy Mapping, POV, and HMW

SAYS (Direct Quotes) <i>"I never know if the form was received."</i>	THINKS (Inferred Beliefs & Doubts) <i>"Will I be billed twice if I click submit again?"</i>
DOES (Observable Workarounds) <i>Takes screenshots, calls customer support.</i>	FEELS (Emotions) <i>Anxious, confused, hesitant.</i>
PAINS: Financial uncertainty GAINS: Clear confirmation	

POV and HMW Formulation

POV = [User Profile] + [Core Need] + [Surprising Insight]

HMW Question = "How might we [actionable challenge] without [prescribing exact solution]?"

CHAPTER 3

Unit 3: Ideation and Creative Problem Solving

Unit 3 Essential Summary

- **Ideation:** Transitioning from problem definition to wide candidate solution generation.
- **Golden Rule:** Defer judgment; strictly separate divergent generation from convergent selection.
- **4 Creativity Metrics:** Fluency (volume), Flexibility (diversity of categories), Originality (rarity), Elaboration (level of detail).

3.1 Ideation Toolkits & Associative Techniques

- **Brainstorming Rules (Alex Osborn, 1953):** Defer judgment, go for quantity, encourage wild ideas, build on others using “Yes, and...”.
- **Brainwriting (6-3-5):** 6 participants write 3 ideas per sheet for 5 minutes, rotating sheets silently to eliminate dominant speaker bias (yields 108 ideas in 30 mins).
- **SCAMPER Framework:**
 - **S — Substitute:** Replace materials or actors (paper ticket → QR code).
 - **C — Combine:** Merge distinct tools (phone + camera = smartphone).
 - **A — Adapt:** Borrow from other domains (Kingfisher beak → bullet train nose).
 - **M — Modify / Magnify / Minify:** Alter size, speed, scale (enlarge emergency stop button).
 - **P — Put to Another Use:** Repurpose assets (shipping containers → mobile clinics).
 - **E — Eliminate:** Remove redundant complexity (remove physical keyboard).
 - **R — Reverse / Rearrange:** Invert roles/sequence (Uber drivers come to customer).
- **Reverse Thinking (Inversion / Worst Idea):** Brainstorm ways to make the problem disastrous, then invert findings into robust design features.

3.2 Organizing, Screening & Prioritization

Weighted Idea Screening Matrix Formula

$$T_i = \sum_{j=1}^n (w_j \times s_{ij}) \quad \text{where} \quad \sum w_j = 1.0, \quad s_{ij} \in [1, 5]$$

Example: Desirability ($w = 0.5, s = 4$) + Feasibility ($w = 0.3, s = 5$) + Viability ($w = 0.2, s = 4$)
 $= 2.0 + 1.5 + 0.8 = 4.3$.

- **Impact-Effort 2x2 Matrix:**
 - ***High Impact, Low Effort:** ****Quick Wins**** (Implement first).
 - ***High Impact, High Effort:** ****Major Projects / Strategic Bets**** (Plan carefully).

- *Low Impact, Low Effort:* **Fill-ins** (Do if surplus time permits).
- *Low Impact, High Effort:* **Thankless Tasks / Money Pits** (Drop/Avoid).
- **Dot Voting:** Democratic, visual heat-mapping of team preferences using colored adhesive dots.

CHAPTER 4

Unit 4: Product Design and Prototyping

Unit 4 Essential Summary

- **Prototyping Philosophy:** *“Build to think, test to learn”*. Fail early and cheaply to reduce risk.
- **Riskiest Assumptions First:** Always test whether users need and trust the core value before polishing UI colors.
- **Traceability:** Connect every wireframe element directly back to a validated user requirement.

4.1 Fidelity Comparison & Paper Prototyping

Dimension	Low-Fidelity (Lo-Fi)	High-Fidelity (Hi-Fi)
Materials	Paper, cardboard, sticky notes, sketches.	Clickable Figma mockups, coded frontend.
Time & Cost	Minutes to hours; zero monetary cost.	Days to weeks; requires technical tools.
Testing Focus	Information architecture, layout, user flows.	Visual hierarchy, micro-interactions, metrics.
Feedback	Honest, conceptual, structural critique.	Detailed aesthetic and UI styling comments.

Table 4.1: Lo-Fi vs. Hi-Fi Prototype Trade-Offs.

- **Paper Prototyping Roles:**
 - *User:* Interacts by pointing, tapping, and typing with a pen.
 - *“Computer” (Facilitator):* Silently swaps paper screens and popups.
 - *Observer:* Silently records task time, hesitation, and errors.
- **Storyboarding:** 4-frame comic-strip narrative showing user problem trigger → discovery → interaction → emotional resolution in context.

4.2 Wireframing Conventions & User Flows

- **Grayscale Restraint:** Use only shades of gray to focus on structure, layout, and hierarchy without color distraction.
- **UI Control Rules:**
 - **Radio Buttons (☉):** Exactly ONE selection from a mutually exclusive list.
 - **Checkboxes (☒):** Zero, one, or MULTIPLE independent selections.
 - **Toggle Switch:** Immediate on/off binary setting.
 - **Image Placeholder:** Crossed rectangle (☒).
- **User Flowchart Symbols:**

- *Oval (Terminator):* Start / End of task.
- *Rectangle:* Screen, Step, or System Action.
- *Diamond:* Conditional decision point with branching logic (Yes/No).

4.3 Minimum Viable Product (MVP)

MVP Core Definition & Forms

An MVP is the smallest coherent release that delivers **complete end-to-end value** and enables maximum validated learning.

- **Landing Page MVP:** Web page measuring click-through demand before writing code.
- **Concierge MVP:** Delivering service manually behind the scenes to study user behavior.
- **Wizard of Oz MVP:** Appears automated on front-end, but operated manually by humans in backend.

CHAPTER 5

Unit 5: Testing, Validation and Customer Experience

Unit 5 Essential Summary

- **Usability Testing:** Evaluates if users can use the interface effectively and efficiently (*“Did we build it right?”*).
- **Design Validation:** Evaluates if the product solves the real core human problem (*“Did we build the right thing?”*).
- **“Show, Don’t Tell” Doctrine:** Put the prototype in front of the user; observe natural behavior without pitching or explaining.

5.1 Testing Conduct & Feedback Capture Matrix

- **Neutral Facilitation:** Reassure user (*“We are testing the product, not you”*); use non-leading neutral prompts (*“What did you expect to happen?”*); utilize the Think-Aloud Protocol.
- **Goal-Oriented Task Scenarios:** Provide realistic context, constraints, and end goals without revealing button names or navigation sequences.
- **Feedback Capture Matrix:**
 - **Likes (+):** Elements users praised or found intuitive.
 - **Wishes (Δ):** Constructive criticisms and unmet expectations.
 - **Questions (?):** Confusions and queries raised during testing.
 - **Ideas (!):** New feature suggestions sparked during sessions.

5.2 Usability vs. Customer Experience (CX) & Journey Mapping

Dimension	Usability (UI/UX)	Customer Experience (CX)
Scope	Interface-specific task interaction.	Holistic end-to-end relationship across all channels.
Timeframe	Minutes of active product usage.	Entire customer lifecycle (discovery to off-boarding).
Touchpoints	Screen buttons, menus, mobile app.	Human (couriers), Digital (apps), Physical (packaging).
Key Metrics	Task Completion Rate, Time on Task, SEQ.	Net Promoter Score (NPS), CSAT, Retention Rate.

Table 5.1: Usability vs. Customer Experience (CX) Comparison.

5.3 Digital Accessibility: WCAG POUR Principles

The 4 WCAG Principles

1. **Perceivable (P):** Provide text alternatives (`alt-text`) for non-text media; maintain minimum contrast ratio: **4.5:1** for normal body text, **3:1** for large text.
2. **Operable (O):** Full keyboard navigation without requiring mouse gestures; no keyboard traps.
3. **Understandable (U):** Predictable navigation, inline form validation, and clear error recovery.
4. **Robust (R):** Compatibility across browsers and assistive screen readers.

CHAPTER 6

Unit 6: Innovation Project, Re-Design and Product Presentation

Unit 6 Essential Summary

- **Innovation Lifecycle:** Ongoing empirical loop of building, testing, learning, and refactoring.
- **Refactoring vs. Re-Design:** Refactoring improves internal quality without changing behavior; Re-design fundamentally overhauls core architecture.
- **NABC Model:** Pitching framework covering Need, Approach, Benefits per costs, and Competition.

6.1 Iteration, Prioritization & Pivots

Feedback Priority Formula & PDCA Cycle

$$P = F \times S \times C$$

- $F \in [1, 5]$: Frequency | $S \in [1, 5]$: Severity | $C \in [0.1, 1.0]$: Confidence.
- *Severe safety/task blockers ($S = 5$) outrank common cosmetic tweaks ($S = 1$).*
- **PDCA Continuous Improvement Loop: Plan → Do → Check → Act.**

- **Regression Testing:** Retesting previously working features after any code or design change to ensure zero new defects were introduced.
- **Strategic Pivot:** Course correction changing target audience, channel, or value proposition while preserving validated past learning (e.g., YouTube video dating → universal video sharing).

6.2 Product Presentation Frameworks

6.2.1 Audience Calibration

- **Users:** Highlight personal convenience, comfort, and direct problem relief.
- **Engineers:** Highlight technical feasibility, component specs, latency, and safety.
- **Investors / Executives:** Highlight business viability, unit economics, scalability, ROI, and NABC.

6.2.2 The NABC Framework

- **N — Need:** Specific, quantitative, validated user pain point backed by research.
- **A — Approach:** Unique, feasible technology mechanism or service model.
- **B — Benefits per Costs:** Measurable net value delivered relative to user cost/friction.
- **C — Competition / Alternatives:** Clear relative advantage over existing competitors and workarounds.

6.2.3 The Elevator Pitch Template (30–60 Seconds)

“For [target user] who experiences [validated problem], our [product name] is a [product category] that delivers [core benefit]. Unlike [existing competitor/workaround], our solution [key differentiator]. User testing shows [empirical metric evidence]. We seek [actionable call-to-action].”